

# Yuan YIN 银元

AI Researcher · Model adaptation, learned dynamics, autonomous driving

 [yuan-yin.github.io](https://github.com/yuan-yin) ·  [Google Scholar](https://scholar.google.com/citations?user=yuan-yin) ·  [yuan-yin-nn](https://www.linkedin.com/in/yuan-yin-nn) ·  [yuan-yin](mailto:yuan-yin@valeo.ai) ·  Paris, France



## Profile

AI researcher working on **parameter-efficient adaptation of pretrained vision models** and **learned physical dynamics**, applied to autonomous driving. My PhD established hybrid physics/deep-learning modelling and out-of-distribution generalisation for dynamical systems; I now transfer that work to autonomous driving at an open industrial research lab, where results are published rather than kept internal. I favour methods that exploit the structure of a problem — physical priors, geometry, pretrained representations — and I work on integrating deep learning into existing non-ML systems.

## Experience

**Valeo.ai** *Paris, France*

► **AI Researcher** *Dec 2024 – Present*

Perception, motion forecasting and decision making for autonomous driving; adapting large pretrained models and learning at scale

► **Postdoctoral AI Researcher** *Apr 2024 – Nov 2024*

Corner-case generation for learning robust self-driving cars

**Sorbonne Université ISIR, MLIA** *Paris, France*

► **Postdoctoral Researcher** *Jul 2023 – Dec 2023*

Supervised ongoing research projects; authored a tutorial on physics-aware deep learning

► **PhD Student & Teaching Assistant** *Oct 2019 – Jun 2023*

Advisors: Patrick Gallinari & Nicolas Baskiotis  
*Physics-aware deep learning for dynamical systems:* hybrid physics/DL modelling, out-of-distribution generalisation, continuous-time dynamics

► **Research Intern, Deep Learning** *Feb 2019 – Sep 2019*

Imputing spatiotemporal data with generative models

**Inria Paris** *Research Intern, NLP* *2018*

**Beihang Univ.** *Research Intern, Vision* *2015–16*

## Education

**Sorbonne Université** fka UPMC (Paris-6) *Paris, France*

► **PhD, Machine Learning & Deep Learning** *2023*

► **MSc2 DAC** *Master Data Science Paris* *2019*

**Université Paris Cité** fka Paris-Diderot (Paris-7) *Paris, France*

► **MSc1 MPRI** *Parisian Research Master in CS* *2018*

► Univ. Diploma, French Language & Civilization *2017*

**Beihang University** *Beijing, China*

► **BSc, Computer Science** *2016*

## Technical Skills

**ML / DL** PyTorch, JAX, Hugging Face, multi-GPU & multi-node training

**Coding** Python, C/C++, Java, OCaml, Matlab,  $\text{\LaTeX}$   
**Infra** Linux, Slurm, Docker, Git, Weights & Biases

## Languages

**French** Bilingual *last exam ▶ C1 (2017)*

**English** Full Professional *last exam ▶ B2 (2015)*

**Mandarin** Native

## Awards & Honours

**Accessit** for the 2024 AI Thesis Prize of the [French Association for Artificial Intelligence \(AFIA\)](#)

**Best Paper & Oral**, NeurIPS 2025 CCFM Workshop

**Top Reviewer**, NeurIPS 2023

## Service & Teaching

**Conference reviewing** — NeurIPS 2021–26, ICLR 2023–26, ICML 2022–26, CVPR 2025–26, ICRA 2026, ECML-PKDD 2021, ACM-MM 2021

**Workshop reviewing** — CCFM @ NeurIPS 2025, ML4PS @ NeurIPS 2022–24, Physics4ML @ ICLR 2023, SynS&ML @ ICML 2023, ROAM @ ECCV 2024

**Teaching** — Sorbonne Université, Engineering Dept. (UFR 919), 2019–22, in French. Undergraduate: C programming (L1), algorithmics (L2), probability (L3). Graduate: ML research methodology (M2)

## Selected Publications

**Conference & journal papers** *\* equal contribution*

· E. Kirby, A. Boulch, Y.-H. Xu, **Y. Yin**, G. Puy, É. Zablocki, A. Bursuc, S. Gidaris, R. Marlet, F. Bartoccioni, A.-Q. Cao, N. Samet, T.-H. Vu, and M. Cord. Driving on registers. In **CVPR 2026**.

· Y.-H. Xu\*, **Y. Yin\***, T.-H. Vu, A. Boulch, É. Zablocki, and M. Cord. PPT:

Pre-training with pseudo-labeled trajectories for motion forecasting. In **ICRA 2026**.

- **Y. Yin**, S. Venkataramanan, T.-H. Vu, A. Bursuc, and M. Cord. IPA: An information-reconstructive input projection framework for efficient foundation model adaptation. **TMLR**, 2025.

**Best Paper & Oral, NeurIPS 2025 CCFM WS**

- L. Le Boudec, E. de Bézenac, L. Serrano, R. D. Regueiro-Espino, **Y. Yin**, and P. Gallinari. Learning a neural solver for parametric PDE to enhance physics-informed methods. In **ICLR 2025**.
- A. Kassai Koupaï, J. Mifsut-Benet, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. GEPS: Boosting generalization in parametric PDE neural solvers through adaptive conditioning. In **NeurIPS 2024**.
- E. Le Naour, L. Serrano, L. Migus, **Y. Yin**, G. Agoua, N. Baskiotis, P. Gallinari, and V. Guigue. Time series continuous modeling for imputation and forecasting with implicit neural representations. **TMLR**, 2024.
- L. Serrano, L. Le Boudec, A. Kassai Koupaï, T. X. Wang, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. Operator learning with neural fields: Tackling PDEs on general geometries. In **NeurIPS 2023**.
- **Y. Yin**<sup>\*</sup>, M. Kirchmeyer<sup>\*</sup>, J.-Y. Franceschi<sup>\*</sup>, A. Rakotomamonjy, and P. Gallinari. Continuous PDE dynamics forecasting with implicit neural representations. In **ICLR 2023**. **Spotlight**
- M. Kirchmeyer<sup>\*</sup>, **Y. Yin**<sup>\*</sup>, J. Donà, N. Baskiotis, A. Rakotomamonjy, and P. Gallinari. Generalizing to new physical systems via context-informed dynamics model. In **ICML 2022**. **Spotlight**
- **Y. Yin**, I. Ayed, E. de Bézenac, N. Baskiotis, and P. Gallinari. LEADS: Learning dynamical systems that generalize across environments. In **NeurIPS 2021**.
- **Y. Yin**<sup>\*</sup>, V. Le Guen<sup>\*</sup>, J. Donà<sup>\*</sup>, E. de Bézenac<sup>\*</sup>, I. Ayed<sup>\*</sup>, N. Thome, and P. Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In **ICLR 2021**.

**Oral; also in J. Stat. Mech. Theory Exp.**

- D. Huang, R.K. Zhang, **Y. Yin**, Y.D. Wang, and Y.H. Wang. Local feature approach to dorsal hand vein recognition by centroid-based circular key-point grid and fine-grained matching. **Image Vis. Comput.**, 2017.

## Workshop papers

- **Y. Yin**, P. Khayatan, É. Zablocki, A. Boulch, and M. Cord. ReGentS: Real-world safety-critical driving scenario generation made stable.

In **ECCV 2024 Workshop on W-CODA**.

- L. Serrano, L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. INFINITY: Neural field modeling for reynolds-averaged navier-stokes equations. In **ICML 2023 Workshop on SynS & ML**.
- L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. Multi-scale physical representations for approximating PDE solutions with graph neural operators. In **ICLR 2022 Workshop on GTRL**.
- **Y. Yin**, A. Pajot, E. De Bézenac, and P. Gallinari. Unsupervised inpainting for occluded sea surface temperature sequences. In **CI 2019**.

## Preprints

- **Y. Yin**, E. Ramzi, M. Lafon, V. Charraut, V. Bares, Y.-H. Xu, É. Zablocki, A. Boulch, T. Buhet, A. Bursuc, and M. Cord. Pictura: Perspective-view self-play at scale for driving. **arXiv** 2607.26005, 2026.
- Y.-H. Xu, É. Zablocki, **Y. Yin**, E. Ramzi, E. Kirby, A. Boulch, and M. Cord. TOAD: Test-time trajectory optimization for autonomous driving. **arXiv** 2606.07170, 2026.

*Full list of publications on [Google Scholar](#)*

## Selected Talks

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|----------------------------------------------------------------------------|----------|
| Oral, NeurIPS 2025 CCFM Workshop                                           | Dec 2025 |
| Poster, ECCV 2024                                                          | Sep 2024 |
| <a href="#">Workshop on Mathematical Foundations of AI, DATAIA-SCAI</a>    | Jan 2024 |
| Tutorial, <a href="#">ECML-PKDD 2023</a>                                   | Sep 2023 |
| Seminar, <a href="#">Signal Processing Lab (LTS4), EPFL</a>                | May 2023 |
| Spotlight, <a href="#">ICLR 2023</a>                                       | May 2023 |
| <a href="#">AI4Science Talks, Univ. of Stuttgart &amp; NEC Labs Europe</a> | Apr 2023 |
| Seminar, <a href="#">Criteo AI Lab</a>                                     | Nov 2022 |
| Spotlight, <a href="#">ICML 2022</a>                                       | Jul 2022 |

*Full list of talks on [my website](#)*